Thermodynamic description of the Pb–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

Synthetic document generated by the ORCHID pipeline benchmark; not a real publication. CC BY 4.0. Derived from the SGTE binary dataset collection (B. Hallstedt, Calphad 89 (2025) 102833, CC BY 4.0) and the original assessment cited in the dataset.

Abstract

This synthetic document describes the Pb–Sn binary system as calculated with pycalphad from the SGTE dataset PbSn-17Gie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Pb) + (Sn), lies at 182.7 °C.

1. Introduction

The Pb–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of W. Gierlotka, Y.-C. Tung, J. Phase Equilib. Diffus., 38, 814-28(2017). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Pb reproduced by the dataset is 327.5 °C. Pure Sn melts at 231.9 °C.

2. Phase diagram and invariant reactions

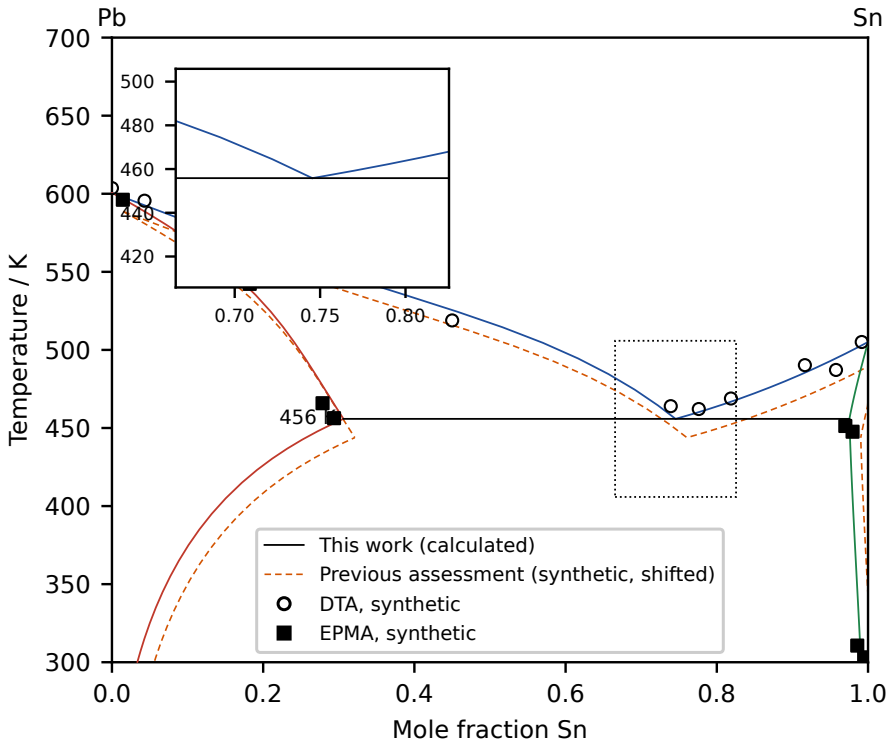


Fig. 1. Calculated Pb–Sn phase diagram with synthetic experimental-style points and a dashed, deliberately shifted second assessment (inset: enlarged view near the lowest invariant).

At 182.7 °C the eutectic L → (Pb) + (Sn) takes place; the L phase then contains 74.5 at.% Sn.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Pb–Sn).

Reaction	Type	T / K		Compositions (x(Sn))		
		this work	exp.	phase 1	phase 2	phase 3
L → (Pb) + (Sn)	eutectic	455.8	464.8 [4]	0.745	0.306	0.975

Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Pb–Sn).

	Structure			Stability range / K		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Sn)	at T / K
(Pb)	cF4	A1	Cu	< 300[a]	600.6	0.000–0.301	460
(Sn)	tI4	A5	β-Sn	< 300[a]	505.1	0.976–1.000	450

[a] Stable at the lowest calculated temperature.

The upper stability limit of (Pb) is 327.5 °C. The solubility of Sn in (Pb) reaches 30.6 at.% at 182.7 °C.

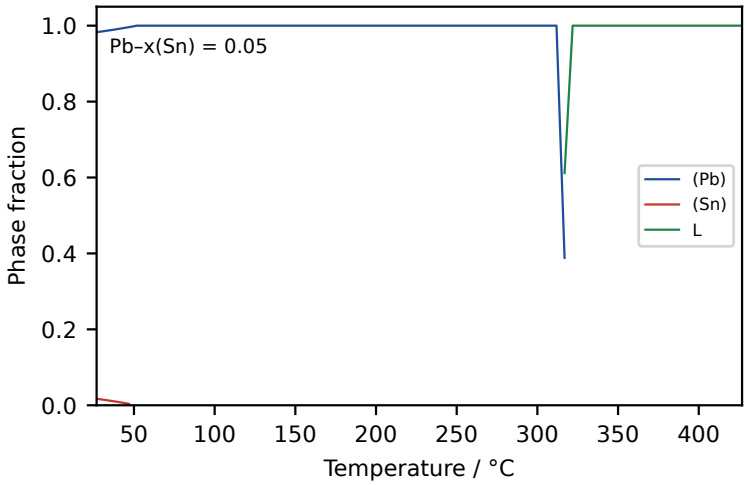


Fig. 2. Calculated equilibrium phase fractions versus temperature for Pb–x(Sn) = 0.05.

4. Thermodynamic properties

At 700 K the integral mixing enthalpy of the liquid has its maximum, 1.29 kJ/mol, at 48.0 at.% Sn. For the equiatomic liquid at 700 K, the Gibbs energy of mixing is −2.47 kJ/mol.

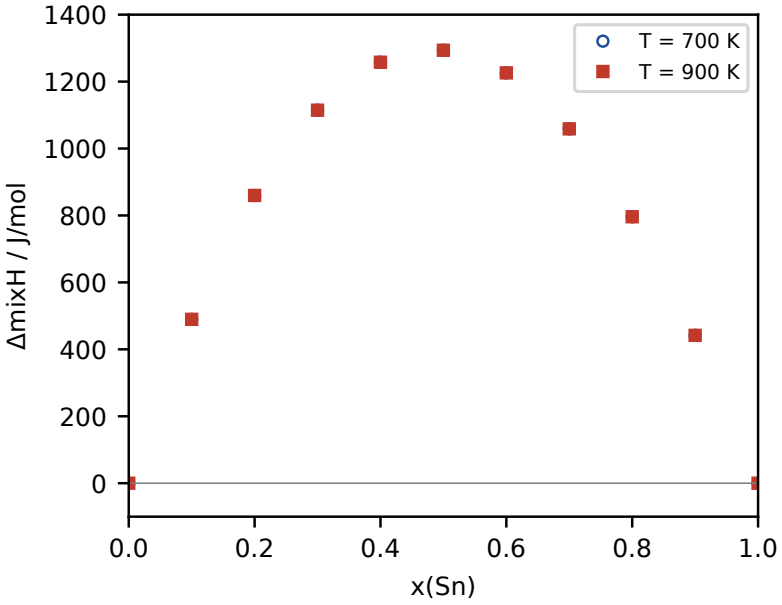


Fig. 3. Calculated mixing enthalpy of the liquid at 700 K and 900 K.

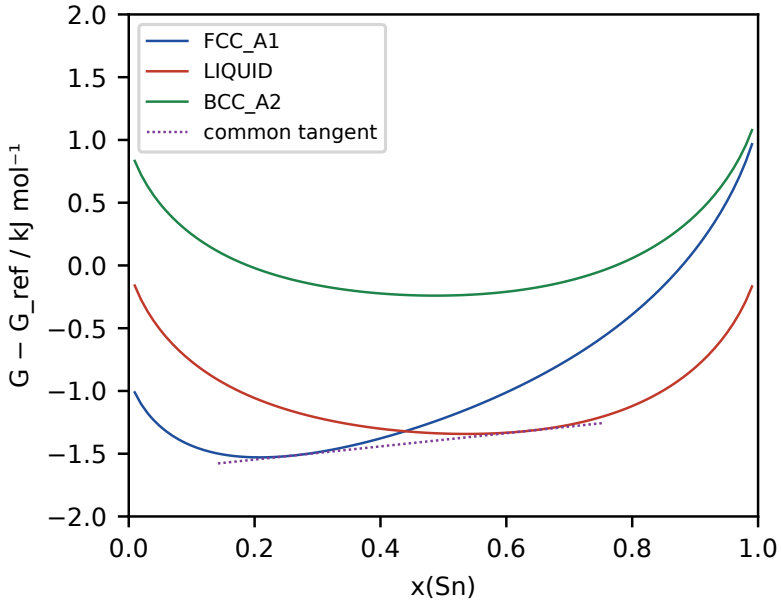


Fig. 4. Calculated Gibbs energy curves at 490 K.

Table 3. Redlich–Kister parameters of the liquid (Pb–Sn).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	5174.58	1.558	0.0
L1	331.69	0.0	0.0

Values as printed in the TDB file.

Table 4. Integral and partial mixing properties of the liquid at 700 K (Pb–Sn).

x(Sn)	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Pb)	a(Sn)	ΔH(Pb) / J mol⁻¹	ΔH(Sn) / J mol⁻¹
0.100	490	-1304	0.9111	0.2459	60	4353
0.300	1115	-2212	0.7784	0.5056	519	2503
0.500	1294	-2468	0.6638	0.6451	1377	1211
0.700	1059	-2267	0.5112	0.7641	2568	412
0.900	442	-1352	0.2326	0.9084	4030	43

Reference states: pure liquid Pb and Sn at 700 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] W. Gierlotka, Y.-C. Tung, J. Phase Equilib. Diffus., 38, 814-28(2017).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.